

Books :

- **Computer Fundamentals**
 - Dr. M. Lutfar Rahman
 - Dr. M. Alamgir Hossain
- **Computer Organization and Architecture**
 - by T K Ghosh and A J Pal
- **Introduction to Computer**
 - by Peter Norton

Syllabus

CSE-1101: Fundamentals of Computers and Computing [2.0 credits, 30 hours lecture]
(Pre-requisite Courses: None)

Introduction to Computers: From a Key Press to Display, Hardware, Software, Operating System, Microprocessor, Memory Overview, File and File System. **Input-Output Devices.** **Application Software:** Basic Text Editor (gedit, Notepad), Document Processing, Spreadsheet, Presentation, Database, Mathematical Analysis, Simulation, Image and Video Editing, Games etc. **Network and Internet:** Networking Concept and Topologies, Network Addresses (MAC, IP and Port), Name vs. IP (role of DNS). **Browser Software:** Examples, URL, Security, Email, Email Address, Email - Client Software, Email Software in the Internet, Network Configuration and Basic Tools (ping, traceroute etc.). **Number System:** Concept of Bit, Electronic Representation of Bits. **Bit- Array:** Byte, Word, Double Word. Binary - to- Decimal Conversion, Binary Arithmetic, Bit- Shifting, Logic Representation (1-Bit, Bit- Array). Hexadecimal Arithmetic up to 32- Bit Array Representation. Conversion between Binary, Hexadecimal and Octal Numbers. **Representation of Characters by Bit-Array:** ASCII and UTF-8. **Character Arithmetic:** Case and Language Mapping and Changing. **Memory:** Introduction to Computer Memory. **System Modeling and Flow Chart.** **Introduction to Programming:** Program Structure, Variables, Constants, I/O, Conditional Statements (If- Else), More about Conditional Statements (Nested If).

Outline

- What is Computer?
- Basic Functions of Computer
- Characteristics of Computer
- Components of a Computer
- Limitations of Computer
- Types of Computer
- History and Generation of Computer

What is a Computer?

- A computer is a programmable electronic device that accepts raw data as input and processes it with a set of instructions (a program) to produce the result as output.
- A computer is designed to execute applications and provides a variety of solutions through integrated hardware and software components.
- It is believed that the Analytical Engine was the first computer which was invented by Charles Babbage in 1837.
- Charles Babbage is also known as the father of the computer.

Basic Functions of Computer

The three fundamental functions of a computer are:

- Input
- Processing and
- Output

Basic Functions of Computer cont..

Input:

- Computers can accept data and instructions from various input devices.
- This includes input from devices like keyboards, mice, touchscreens, scanners, microphones, and more.
- Input devices allow users to communicate with the computer and provide it with data to process.

Processing:

- Once the computer has received data and instructions, it processes this information.
- It performs calculations, executes programs, and manipulates data according to the instructions provided.
- The central processing unit (CPU) is responsible for carrying out these processing tasks.

Basic Functions of Computer cont..

Output:

- After processing the input data, computers generate output, which can take various forms.
- Common output devices include monitors, printers, speakers, and other display or presentation devices.

Characteristics of Computer

- **Speed**—It has fast speed operation of several million operations per second.
- **Accuracy**—Computers perform calculations with 100% accuracy. Errors may occur due to data inconsistency or inaccuracy.
- **Storage**—It is capable of storing large volumes of information.
- **Decision making**—It is capable of decision making according to the supplied information

Components of Computer

- The computer system consists of three main components.
 - Hardware
 - Software
 - Human Resources

Components of Computer - Hardware

- **Hardware :**

- The hardware are physical devices attached with the computer such as central processing unit (CPU), keyboard, monitor, disk drive, printer and other peripherals.

Components of Computer – Software

Software :

- Software consists of the instructions and data that the computers manipulate to perform various tasks.
 - A sequence of instructions is called a program.
 - Generally, software can be either application software or system software.
- **Application software**
 - is a program or collection of programs used to solve a particular application-oriented problem.
 - Examples include editor program, real player, and railway reservation program.
 - **System software**
 - is a program used to manage the entire system and to help in executing various application programs.
 - Operating systems, compilers and device drivers are some of the system software's examples.
 - System-programs are generally machine dependent and is not concerned with specific application program.

Components of Computer – Human Resources

Human Resources :

- It is a manpower and skilled personal (System Analyst, Programmers, Computer Operator etc..) available to perform the operation on computer systems.

Limitations of Computer

- Computers can not think.
- Computers can not do anything without human instructions.
- Computers can not make any adjustment as human being can do.

Types of Computer

Classification based on purposes :

- Special purpose computer and
- General purpose computer

Classification based on signals :

- Analog computer and
- Digital Computer

Classification based on capacity :

Based on capacity (capacity is based on throughput – amount of processing that can be performed in a given amount of time) can be divided into four categories :

- Microcomputers
- Minicomputers
- Mainframe computers and
- Super computers

Types of Computer cont..

Special-Purpose Computer :

- A special-purpose computer, as the name implies, is designed for a specific, dedicated function or set of functions.
- These computers are optimized to excel in particular tasks but may lack the versatility of general-purpose computers.
- **Example :** digital cameras, microwave ovens, washing machines, gaming consoles like the PlayStation or Xbox etc..

General-Purpose Computer:

- A general-purpose computer, often simply referred to as a "computer," is designed to perform a wide range of tasks and functions.
- It is highly versatile and can be programmed to execute a variety of applications and processes.
- **Examples:** Personal computers (desktops, laptops), servers, workstations, and most modern computing devices like smartphones and tablets are general-purpose computers.

Types of Computer cont..

Analog Computer :

- An analog computer represents quantities by physical analogies.
- It represent physical quantities such as distance, velocity, acceleration, temperature, pressure etc. by mechanical or electrical parameters.
- **Example** : analog thermometer, speedometer etc...

Digital Computer:

- Digital computers represent and process data using discrete values or binary code, typically 0s and 1s.
- They use digital logic gates and circuits to perform calculations and execute instructions.
- **Examples:** Personal computers, smart phones, calculator etc..

Types of Computer cont..

Super Computer :

- Supercomputers are the biggest and fastest computers.
- They are designed to process huge amount of data.
- A supercomputer can process trillions of instructions in a second.
- It has thousands of interconnected processors.
- Supercomputers are particularly used in scientific and engineering applications such as weather forecasting, scientific simulations and nuclear energy research.
- The first supercomputer was developed by Roger Cray in 1976.
- **Example :**
 - Summit, an IBM supercomputer at the Oak Ridge National Laboratory, is one of the world's most powerful supercomputers.
 - Fugaku, developed by RIKEN and Fujitsu, is a Japanese supercomputer used for various scientific and engineering applications, including drug discovery and climate modeling.

Types of Computer cont..

Mainframe computer :

- Mainframe computers are designed to support hundreds or thousands of users simultaneously.
- They can support multiple programs at the same time.
- It means they can execute different processes simultaneously.
- These features of mainframe computers make them ideal for big organizations like banking and telecom sectors, which need to manage and process high volume of data.

Minicomputer :

- It is a midsize multiprocessing computer.
- It consists of two or more processors and can support 4 to 200 users at one time.
- Mini computers are used in institutes and departments for tasks such as billing, accounting and inventory management.
- A minicomputer lies between the mainframe and microcomputer as it is smaller than mainframe but larger than a microcomputer.

Types of Computer cont..

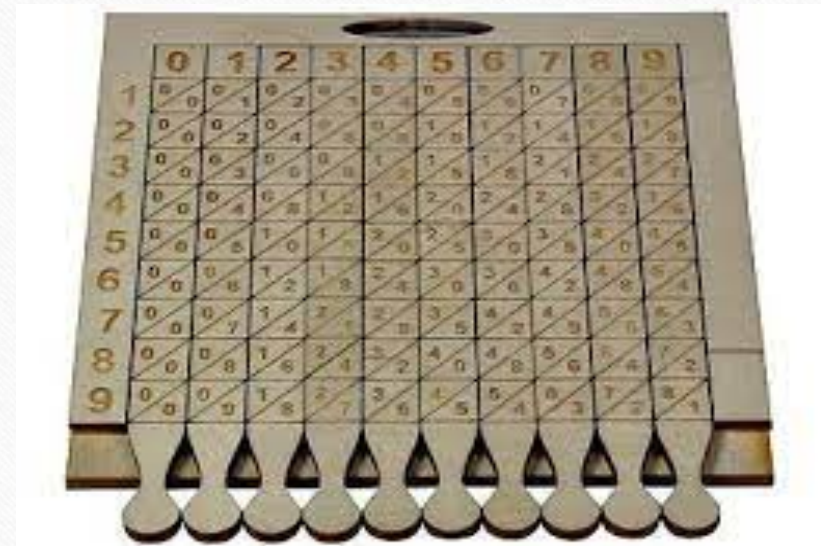
Microcomputer :

- Microcomputer is also known as a personal computer.
- It is a general-purpose computer that is designed for individual use.
- It has a microprocessor as a central processing unit, memory, storage area, input unit and output unit.
- Laptops and desktop computers are examples of microcomputers.
- They are suitable for personal work that may be making an assignment, watching a movie, or at office for office work.

History of Computer

- **Napier's Bones**

- It was a manually-operated calculating device which was invented by John Napier (1550-1617) of Merchiston.
- In this calculating tool, he used 9 different ivory strips or bones marked with numbers to multiply and divide.



History of Computer cont..

Pascaline

- Pascaline is also known as Arithmetic Machine or Adding Machine.
- It was invented between 1642 and 1644 by a French mathematician-philosopher Blaise Pascal.
- It is believed that it was the first mechanical and automatic calculator.



Stepped Reckoner or Leibnitz wheel

- It was developed by a German mathematician-philosopher Gottfried Wilhelm Leibnitz in 1673.
- He improved Pascal's invention to develop this machine.



History of Computer cont..

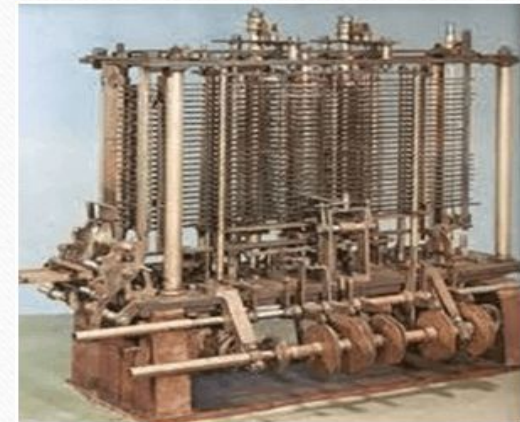
Difference Engine

- In the early 1820s, it was designed by Charles Babbage who is known as "Father of Modern Computer".
- It was a mechanical computer which could perform simple calculations.



Analytical Engine

- This calculating machine was also developed by Charles Babbage in 1830.
- It was a mechanical computer that used punch-cards as input.
- It was capable of solving any mathematical problem and storing information as a permanent memory.



History of Computer cont..

Tabulating Machine

- It was invented in 1890, by Herman Hollerith, an American statistician.
- It was a mechanical tabulator based on punch cards.
- It could tabulate statistics and record or sort data or information.



Differential Analyzer

- It was the first electronic computer introduced in the United States in 1930.
- It was an analog device invented by Vannevar Bush.
- This machine has vacuum tubes to switch electrical signals to perform calculations.
- It could do 25 calculations in few minutes.



History of Computer cont..

Mark I

- The next major changes in the history of computer began in 1937 when Howard Aiken planned to develop a machine that could perform calculations involving large numbers. In 1944, Mark I computer was built as a partnership between IBM and Harvard.
- It was the first programmable digital computer.



Generations of Computer

- A generation of computers refers to the specific improvements in computer technology with time.
- There are five generations of computers which are described below :

First Generation Computers

- The first generation (1946-1959) computers were slow, huge and expensive.
- In these computers, vacuum tubes were used as the basic components of CPU and memory.
- These computers were mainly depended on batch operating system and punch cards.
- Magnetic tape and paper tape were used as output and input devices in this generation;
- Some of the popular first generation computers are;
 - **ENIAC** (Electronic Numerical Integrator and Computer)
 - **EDVAC** (Electronic Discrete Variable Automatic Computer)
 - **UNIVACI**(Universal Automatic Computer)
 - **IBM-701**
 - **IBM-650**

Generations of Computer cont..

Second Generation Computers

- The second generation (1959-1965) was the era of the transistor computers.
- Assembly language and programming languages like COBOL and FORTRAN, and Batch processing and multiprogramming operating systems were used in these computers.
- Some of the popular second generation computers are;
 - **IBM 1620**
 - **IBM 7094**
 - **CDC 1604**
 - **CDC 3600**
 - **UNIVAC 1108**

Generations of Computer cont..

Third Generation Computers

- The third generation computers used integrated circuits (ICs) instead of transistors.
- Also, the high-level programming languages like FORTRAN-II TO IV, COBOL, PASCAL PL/1, ALGOL-68 were used in this generation.
- Some of the popular third generation computers are;
 - **IBM-360 series**
 - **Honeywell-6000 series**
 - **PDP(Personal Data Processor)**
 - **IBM-370/168**
 - **TDC-316**

Generations of Computer cont..

Fourth Generation Computers

- The fourth generation (1971-1980) computers used very large scale integrated (VLSI) circuits; a chip containing millions of transistors and other circuit elements.
- These chips made this generation computers more compact, powerful, fast and affordable.
- The programming languages like C, C++, DBASE were also used in this generation.
- Some of the popular fourth generation computers are;
 - **DEC 10**
 - **STAR 1000**
 - **PDP 11**
 - **CRAY-1(Super Computer)**
 - **CRAY-X-MP(Super Computer)**

Generations of Computer cont..

Fifth Generation Computers

- In fifth generation (1980-till date) computers, the VLSI technology was replaced with ULSI (Ultra Large Scale Integration).
- It made possible the production of microprocessor chips with ten million electronic components.
- This generation computers used parallel processing hardware and AI (Artificial Intelligence) software.
- The programming languages used in this generation were C, C++, Java, .Net, etc.
- Some of the popular fifth generation computers are;
 - **Desktop**
 - **Laptop**
 - **NoteBook**
 - **UltraBook**
 - **ChromeBook**

Thank You

